

Module-IV

Lattices and Boolean algebra

- Partially Ordered Relations
- Lattices as Posets
- Hasse Digram
- Properties of Lattices
- Boolean algebra
- Properties of Boolean Algebra
- Boolean functions

Relation: If A and B are two sets, then relation R from A to B is the subset of the cartesian product $A \times B$.

If a is related to b under the relation R , then we write $a R b$.

Thus $R = \{(x, y) : x \in A, y \in B \text{ and } xRy\}$

The set of first entries of the ordered pairs in a relation is called the **domain of the relation**. The set of second entries of the ordered pairs in a relation is called the **range of the relation**.

Example: If $A = \{2, 3, 4, 5, 6\}$, $B = \{2, 4, 6, 8\}$ and $xRy : x$ divides y , then

$$R = \{(2, 2), (2, 4), (2, 6), (2, 8), (3, 6), (4, 4), (4, 8), (6, 6)\}$$

Domain of $R=\{2,3,4,6\}$,

Range of $R=\{2,4,6,8\}$

Note:

For finite sets A and B with $|A| = m$ and $|B| = n$, there are 2^{mn} relations from A to B .

There are also 2^{mn} relations from B to A .

PROPERTIES OF RELATIONS

Reflexive Relation

Let R be a relation defined in a set A ; then R is reflexive if $a R a$ holds for all $a \in A$, i.e., if $(a, a) \in R$ for all $a \in A$.

Example 1: Let $A = \{a, b, c\}$ and $R = \{(a, a), (b, b), (c, c)\}$ then R is a reflexive relation in A .

Example 2: 'Equality' is a reflexive relation, since an element equals itself.

Symmetric Relation

A relation R defined in set A is said to be 'symmetric' if $b R a$ holds whenever $a R b$ holds for $b \in A$, i.e., R is symmetric in A if

$$(a, b) \in R \Rightarrow (b, a) \in R$$

Example: Let R be relation 'is perpendicular to' in the set of all straight lines, then R is a symmetric relation.

Transitive Relation

A relation R in set A is said to be transitive if

$$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$$

i.e., if $a R b$ and $b R c \Rightarrow a R c$, where $a, b, c \in A$.

Example 1: Let A denote the set of straight lines in a plane and R be a relation in A defined by 'is parallel to' then R is a transitive relation in A .

Example 2: Let $A = \{1, 2, 3\}$ and $R = \{(1, 1), (2, 2), (2, 3), (3, 2), (3, 3)\}$ then R is transitive.

Anti-symmetric Relation

Let R be a relation in a set A , then R is called anti-symmetric.

$$(a, b) \in R, (b, a) \in R \Rightarrow a = b \quad \forall a, b \in R$$

$$\text{i.e. } a R b \text{ and } b R a \Rightarrow a = b$$

Example : Let N denote the set of Natural Numbers R be a relation in N , defined by ' a is a divisor' of b , i.e., $a R b$ if a divides b then R is anti-symmetric since a divides b and b divides $a \Rightarrow a = b$.

Equivalence Relation

A relation R in a set A is said to be an equivalence relation in A , if R is reflexive, symmetric and transitive.

Example :

(i) Let A be the set of all triangle in a plane and let R be a relation in A defined by 'is congruent to', then R is reflexive, symmetric and transitive.

$\therefore R$ is an Equivalence relation in A .

(ii) Let $A = \{a, b, c\}$, and $R = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$ then R is an equivalence relation in A .

Example: Consider the following relations on $\{1,2,3,4\}$:

$$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\},$$

$$R_2 = \{(1,1), (1,2), (2,1)\},$$

$$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\},$$

$$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\},$$

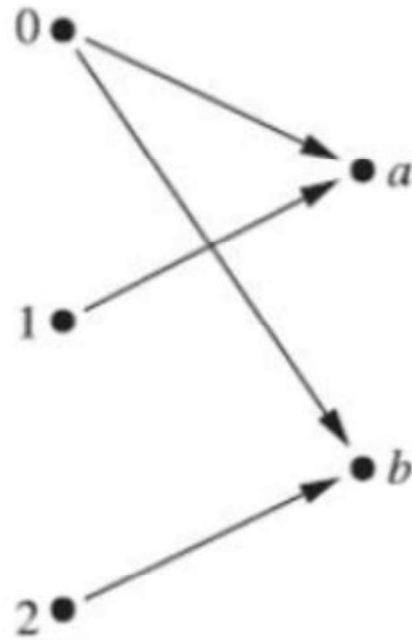
$$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\},$$

$$R_6 = \{(3,4)\}.$$

Which of these relations are reflexive?

Solution: The relations R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) , namely, $(1,1)$, $(2,2)$, $(3,3)$, and $(4,4)$. The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, R_1 , R_2 , R_4 , and R_6 are not reflexive because $(3,3)$ is not in any of these relations.

Let $A = \{0,1,2\}$ and $B = \{a,b\}$. Then $\{(0,a), (0,b), (1,a), (2,b)\}$ is a relation from A to B . This means, for instance, that $0 R a$, but that $1 \not R b$. Relations can be represented graphically,



RELATIONS AND DIGRAPHS

A relation can be represented pictorially by drawing its graph. Let R be a relation on the set $A = \{a_1, a_2, \dots, a_n\}$. The element a_i of A are represented by points (or circles) called nodes (or vertices). If $(a_i, a_j) \in R$ then we connect the vertices a_i and a_j by means of an arc and put an arrow in the direction from a_i to a_j . If $(a_i, a_j) \in R$ and $(a_j, a_i) \in R$ then we draw two arcs between a_i and a_j (sometimes by one arc which starts from node a_i and returns to node a_i (such an arc is called a loop)). When all the nodes corresponding to the ordered pairs in R are connected by arcs with proper arrows, we get a graph of the relation R . If R is reflexive, then there must be a loop at each node in the graph of R . If R is symmetric, then $(a_i, a_j) \in R$ implies $(a_j, a_i) \in R$ and the nodes a_i and a_j will be connected by two arcs (edges) one from a_i to a_j and the other from a_j to a_i .

Example 1: Let $A = \{a, b, d\}$ and R be a relation on A given by

$$R = \{(a, b), (a, d), (b, d), (d, a), (d, d)\}$$

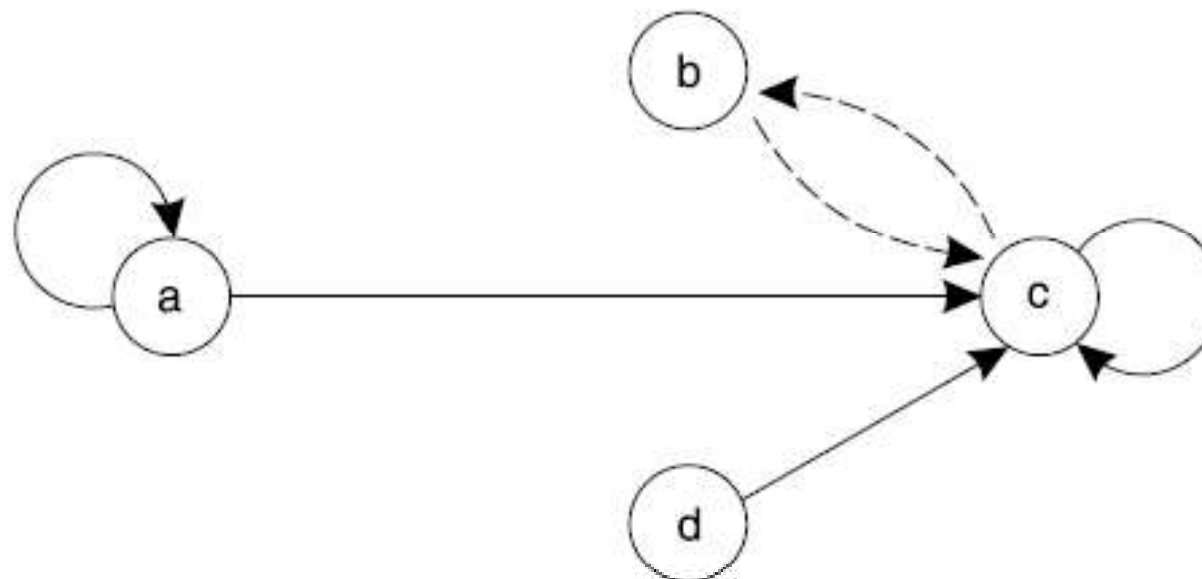
Construct the digraph of R .

Example 2: Let $A = \{1, 2, 3, 4\}$

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 1), (4, 4)\}$$

Construct the digraph of R .

Example 3: Find the relation determined by Fig. 3.9.



Solution: The relation R of the digraph is

$$R = \{(a, a), (a, c), (b, c), (c, b), (c, c), (d, c)\}$$

PARTIAL ORDERING

A relation R on a set S is called a *partial ordering* or *partial order* if it is reflexive, antisymmetric, and transitive. A set S together with a partial ordering R is called a *partially ordered set*, or *poset*, and is denoted by (S, R) . Members of S are called *elements* of the poset.

Example:

Let $A = \{0, 1, 2, 3\}$ and $R = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2), (2, 2), (0, 3), (3, 3)\}$

R is a partial order relation

Comparability

Let R be a partial order on A and $a, b \in A$ whenever aRb or bRa , we say that a and b are comparable otherwise a and b are non-comparable.

TOTALLY ORDERED SET

Definition: A partial order is a total or linear order iff for all x and y in the set either xRy or yRx is true. In a totally ordered set all elements are comparable.

Example 1: Let N be the set of positive integers ordered by divisibility. The elements 5 and 15 are comparable. Since $5/15$ or similarly the elements 7 and 21 are comparable since $7/21$. The positive integers 3 and 5 are non-comparable since neither $3/5$ nor $5/3$. Similarly the integers 5 and 7 are non-comparable.

Example 2: The set N of positive integers with the usual order $\leq$ (less than equal) is a linear order on N . The set $(N, \leq)$ is a totally ordered set.

Hasse Diagrams

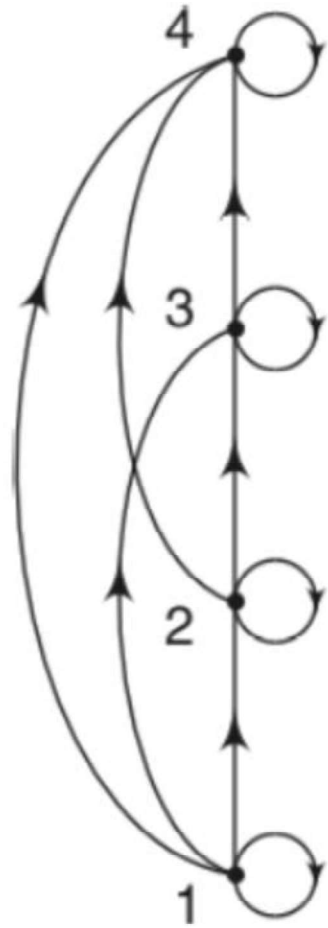
The simplified form of the digraph of a partial ordering on a finite set that contains sufficient information about the partial ordering is called a Hasse diagram, named after the twentieth-century mathematician Helmut Haasse.

The simplification of the digraph as a Hasse diagram is achieved in three ways:

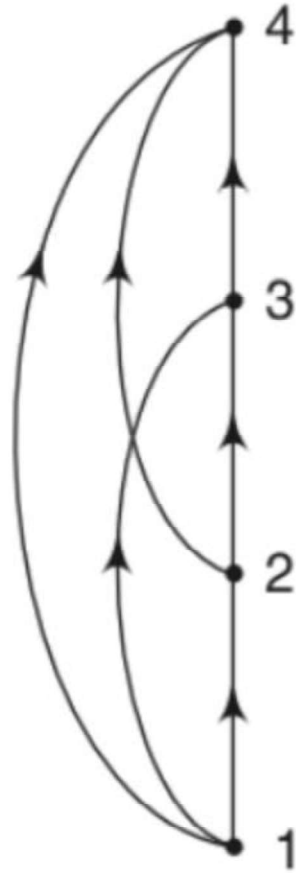
- Since the partial ordering is a reflexive relation, its digraph has loops at all vertices. We need not show these loops since they must be present.
- Since the partial ordering is transitive, we need not show those edges that must be present due to transitivity. For example, if $(1,2)$ and $(2,3)$ are edges in the digraph of a partial ordering, $(1,3)$ will also be an edge due to transitivity. This edge $(1,3)$ need not be shown in the corresponding Hasse diagram.
- If we assume that all edges are directed upward, we need not show the directions of the edges.

Thus the Hasse diagram representing a partial ordering can be obtained from its digraph, by removing all the loops, by removing all edges that are present due to transitivity and by drawing each edge without arrow so that its initial vertex is below its terminal vertex.

Example: let us construct the Hasse diagram for the partial ordering $\{(a, b) \mid a \leq b\}$ on the set $\{1, 2, 3, 4\}$ starting from its digraph.



(a)



(b)



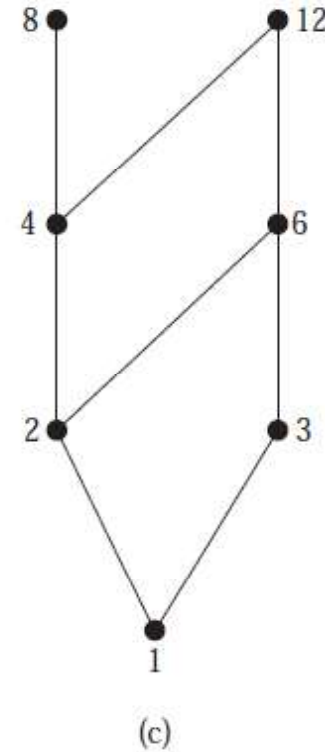
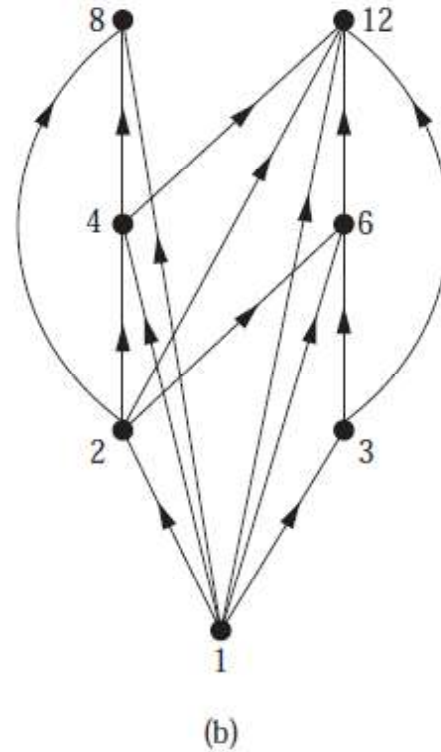
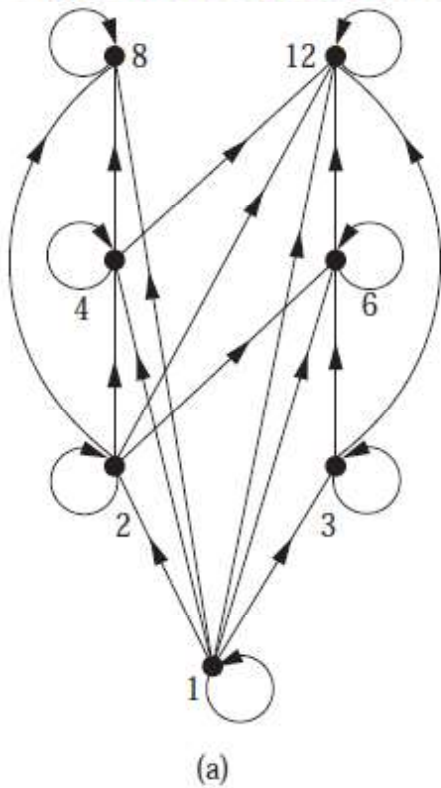
(c)



(d)

Draw the Hasse diagram representing the partial ordering $\{(a, b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.

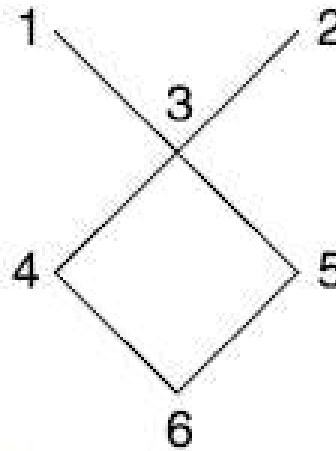
Solution: Begin with the digraph for this partial order, as shown in Figure 3(a). Remove all loops, as shown in Figure 3(b). Then delete all the edges implied by the transitive property. These are $(1, 4)$, $(1, 6)$, $(1, 8)$, $(1, 12)$, $(2, 8)$, $(2, 12)$, and $(3, 12)$. Arrange all edges to point upward, and delete all arrows to obtain the Hasse diagram. The resulting Hasse diagram is shown



UPPER AND LOWER BOUNDS

Sometimes it is possible to find an element that is greater than or equal to all the elements in a subset A of a poset $(S, \preceq)$. If u is an element of S such that $a \preceq u$ for all elements $a \in A$, then u is called an **upper bound** of A . Likewise, there may be an element less than or equal to all the elements in A . If l is an element of S such that $l \preceq a$ for all elements $a \in A$, then l is called a **lower bound** of A .

Example 1: $A = \{1, 2, 3, \dots, 6\}$ be ordered as pictured in



Solution:

If $B = \{4, 5\}$ then

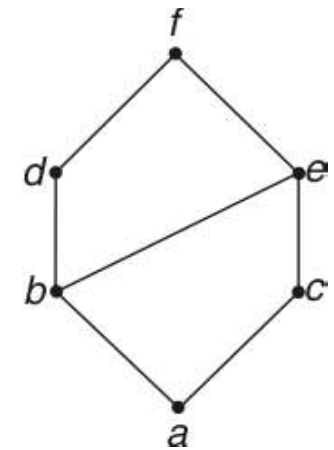
The upper bounds of B are 1, 2, 3

The lower bound of B is 6.

The element x is called the **least upper bound (LUB)** or supremum of the subset A of a poset $\{P, \leq\}$, if x is an upper bound that is less than every other upper bound of A .

Similarly the element y is called the **greatest lower bound (GLB)** or infimum of the subset A of a poset $\{P, \leq\}$, if y is a lower bound that is greater than every other lower bound of A .

Example: let us consider the poset with the Hasse diagram given in Fig



Solution: The upper bounds of the subset $\{a, b, c\}$ are e and f .

[Note: d is not an upper bound, since c is not related to d]

and LUB of $\{a, b, c\}$ is e .

The lower bounds of the subset $\{d, e\}$ are a and b and GLB of $\{d, e\}$ is b .

c is not a lower bound, since c is not related to d .

LATTICES

Definition:

A lattice is a partially ordered set $\{L, \leq\}$ in which every pair of elements $a, b \in L$ has a greatest lower bound and least upper bound.

The greatest lower bound of a subset $\{a, b\} \in L$ will be denoted by $a * b$ and the least upper bound by $a \oplus b$.

Join or sum: The LUB of a subset $\{a, b\} \subseteq L$ is denoted by $a \oplus b$ (or $a \vee b$ or $a + b$) and is called the join or sum of a and b .

Meet or product: The GLB of a subset $\{a, b\} \subseteq L$ is denoted by $a * b$ (or $a \cdot b$ or $a \wedge b$) is called the meet or product of a and b .

A totally ordered set is trivially a lattice, but not all partially ordered sets are lattices

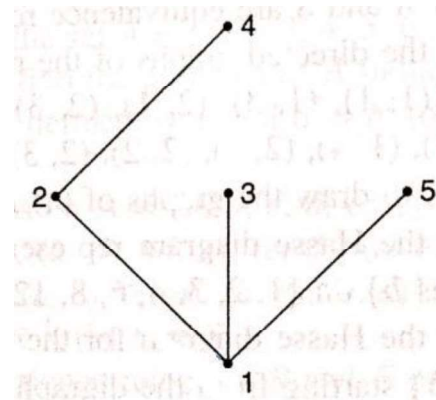
Example: $(\{1,2,4,8\}, |)$, where $|$ means 'divisor of'. The hasse diagram

LUB = 8, GLB = 1

So, it is a lattice.

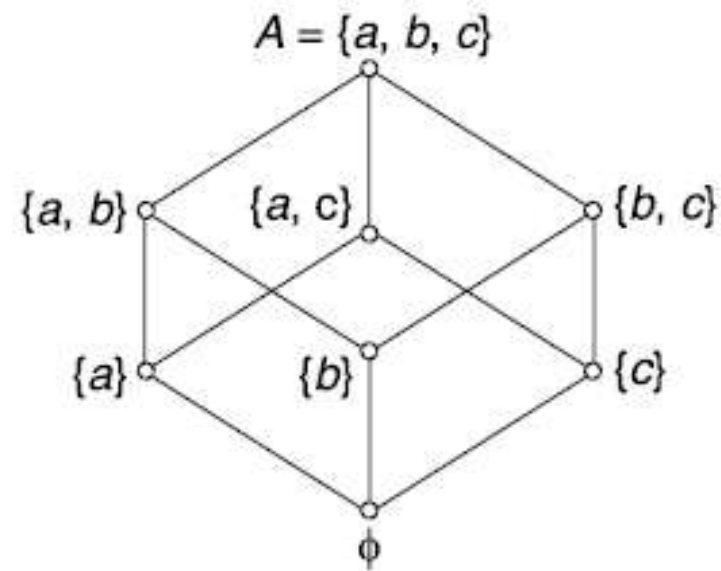


Example: $(\{1,2,3,4,5\}, |)$ It is not a lattice, since LUB of the pair (2, 3) and (3, 5) do not exist.



Example: In the case of power set $P(S)$ of any set S , $(P(S), \subseteq)$ is a lattice

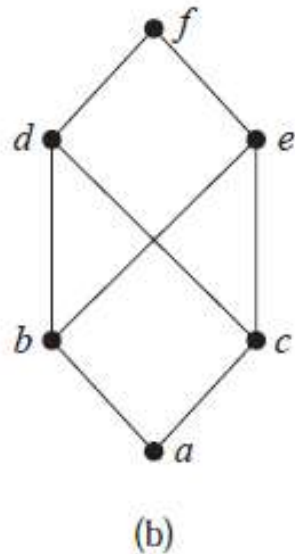
Here $LUB = A \cup B$ and $GLB = A \cap B$, where A and B are any subsets of $P(S)$.



Example: Is the poset $(\mathbb{Z}^+, |)$ a lattice?

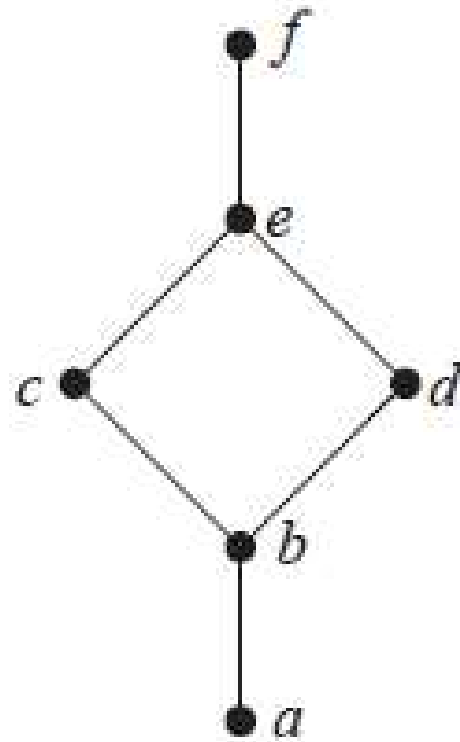
Solution: Let $a, b \in \mathbb{Z}^+$, now, LUB of these two integers is the LCM (Least Common multiple) and GLB is the GCD (Greatest Common Divisor).

Example:



It is not a lattice since the pair (b, c) have no least upper bound.

Example:



It is a lattice since every pair of elements has LUB and GLB.

Properties of Lattice:

We shall first list some of the properties of the two binary operations of meet and join denoted by $*$ and $\oplus$ on a lattice $(L, \leq)$.

For any $a, b, c \in L$, we have

Idempotent law

$$(L - 1) \quad a * a = a,$$

$$(L - 1)' \quad a \oplus a = a$$

Commutative Law

$$(L - 2) \quad a * b = b * a,$$

$$(L - 2)' \quad a \oplus b = b \oplus a$$

Associativity

$$(L - 3) \quad (a * b) * c = a * (b * c),$$

$$(L - 3)' \quad (a \oplus b) \oplus c = a \oplus (b \oplus c),$$

Absorption Law

$$(L - 4) \quad a * (a \oplus b) = a$$

$$(L - 4)' \quad a \oplus (a * b) = a$$

Lattices and Algebraic systems:

Definition:

A lattice is an algebraic system $\langle L, *, \oplus \rangle$ with two binary operations $*$ and $\oplus$ on L which are both (1) commutative and (2) associative and (3) satisfy the absorption laws. In other words the operations $*$ and $\oplus$ satisfy the identities (L-2) to (L-4) and (L-2)' to (L-4)'.

Sub lattices

Definition:

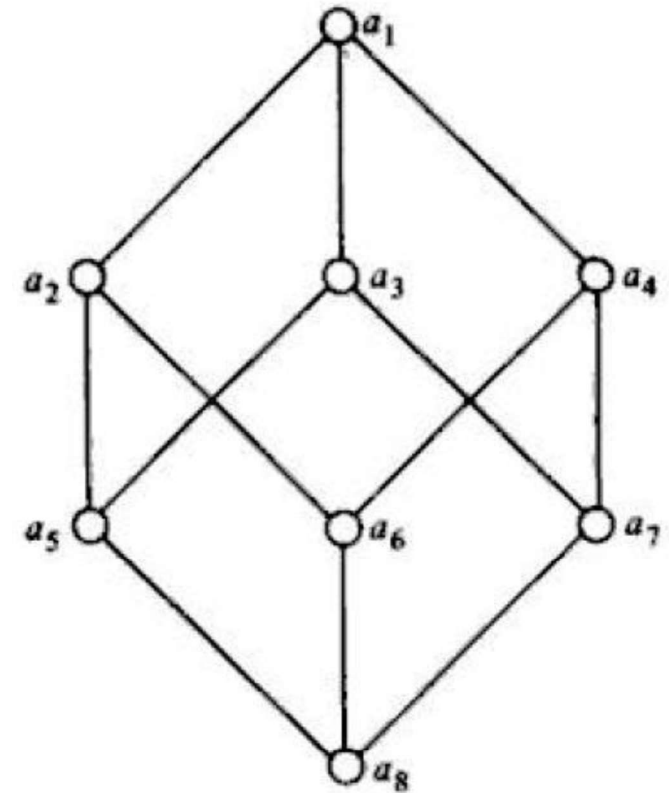
Let $\langle L, *, \oplus \rangle$ be a lattice and let S be a subset of L . The algebra $\langle S, *, \oplus \rangle$ is a sub-lattice of $\langle L, *, \oplus \rangle$ iff S is closed under both operations $*$ and $\oplus$.

Example 1:

Let $\langle L, \leq \rangle$ be a lattice in which $L = \{a_1, a_2, \dots, a_8\}$ and S_1, S_2 and S_3 be the subsets of L given by $S_1 = \{a_1, a_2, a_4, a_6\}$, $S_2 = \{a_3, a_5, a_7, a_8\}$ and $S_3 = \{a_1, a_2, a_4, a_8\}$.

Observe that $\langle S_1, \leq \rangle$ and $\langle S_2, \leq \rangle$ are sub lattices of $\langle L, \leq \rangle$

But $\langle S_3, \leq \rangle$ is not a sub lattices , because $a_2, a_4 \in S_3$ but $a_2 * a_4 = a_6 \notin S_3$ note that $\langle S_3, \leq \rangle$ is a lattices.



Example 2: The lattices of divisor of any positive integer n denoted by $\langle S_n, | \rangle$ is a sub lattices of $\langle \mathbb{Z}^+, | \rangle$.

Some special lattices:

Definition:

A lattice is called **complete** if each of its non-empty subsets has a least upper bound and a greatest lower bound.

Definition:

A lattice $(L, *, \oplus)$ is said to be **bounded** if it has a greatest element and a least element. The greatest and least elements are denoted by 1 and 0 respectively.

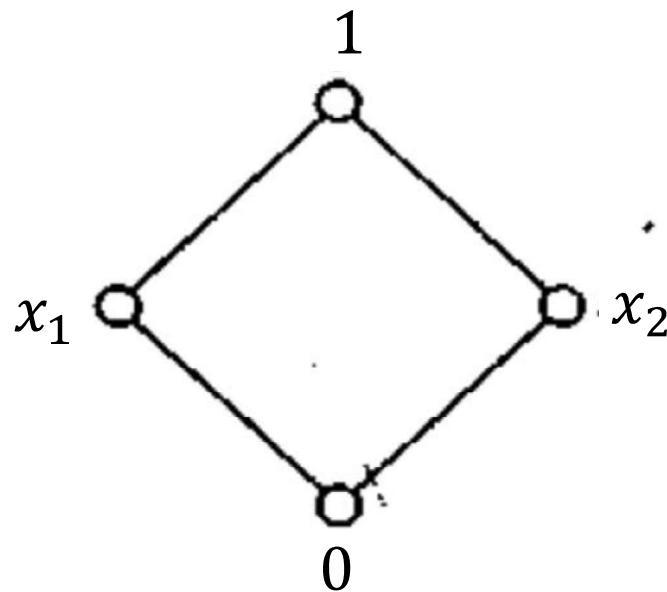
Definition:

In a bounded lattice $\langle L, *, \oplus, 0, 1 \rangle$ an element $b \in L$ is called a **complement of an element $a \in L$** if $a * b = 0$ & $a \oplus b = 1$.

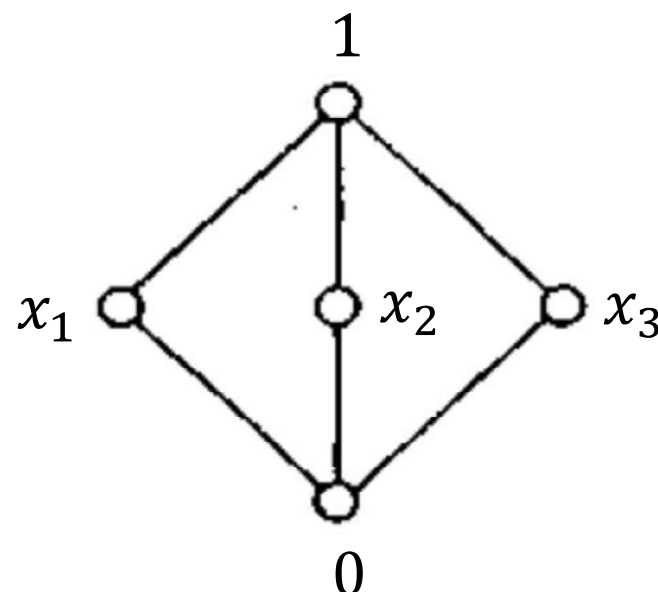
Definition:

A lattice $\langle L, *, \oplus, 0, 1 \rangle$ is said to be **complemented lattices** if every element of L has at least one complement.

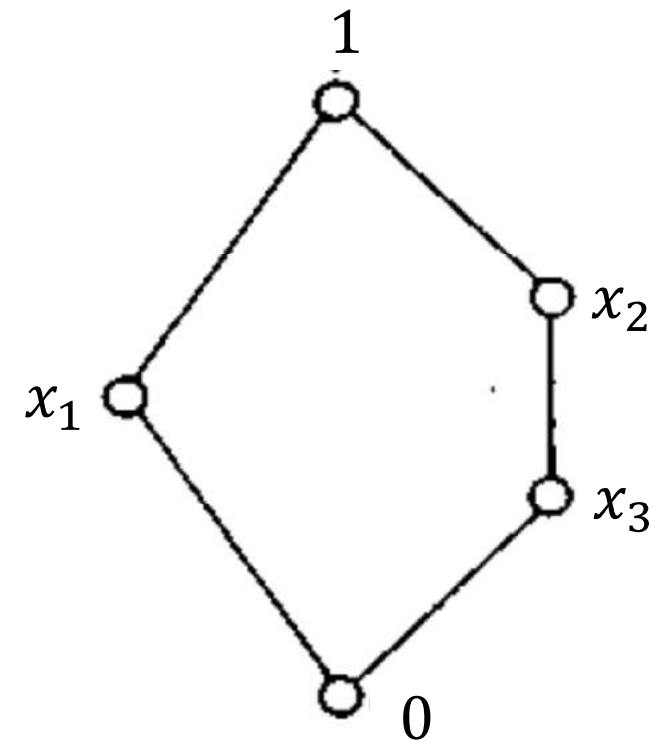
In fig some lattices are shown and the complement of some of the elements are noted below the diagram.



Complement of
 x_1 is x_2

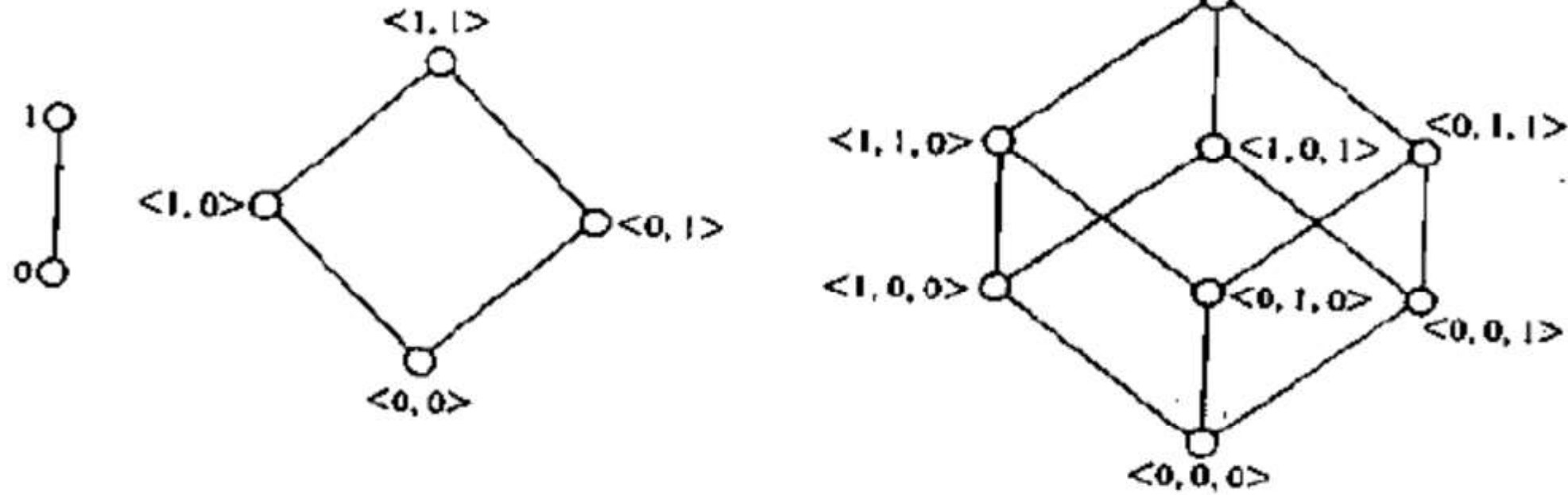


Complements of x_1 are
 x_2, x_3
Complements of x_2 are
 x_1, x_3



Complements of x_1 are x_2, x_3

let $L = \{0,1\}$ and the lattices $\langle L, \leq \rangle$ be as shown in fig.



The lattices $\langle L^2, \leq_2 \rangle$, $\langle L^3, \leq_3 \rangle$, are shown in fig .
In general of $\langle L^n, \leq_n \rangle$, is an n cube.

Example:

Let $\langle L^n, \leq_n \rangle$ be the lattice of n- tuples of 0 and 1 . This is a complemented lattice in which every element has a unique complement.

The complement of an element of L^n can obtained by interchanging 1 by 0 and 0 by 1 in the n- tuples representing the element. As a special case let $n=3$. The bounds of $\langle L^n, \leq_n \rangle$, $\langle 0,0,0 \rangle$ and $\langle 1,1,1 \rangle$ the complement of $\langle 1,0,1 \rangle$ is $\langle 0,1,0 \rangle$

Definition:

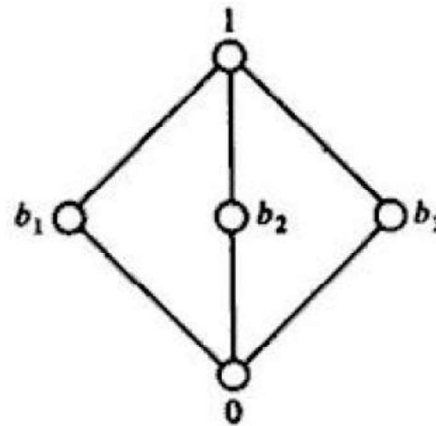
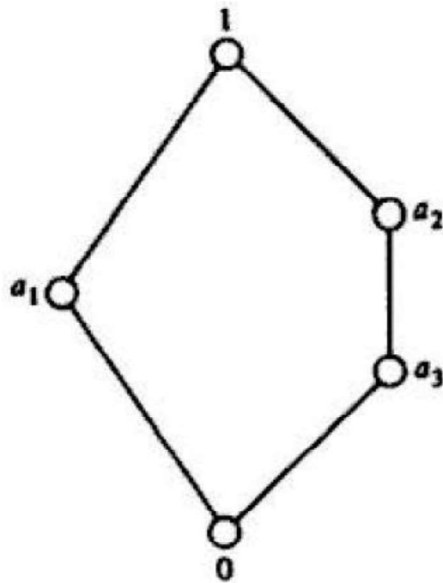
A lattice $\langle L, *, \oplus \rangle$ is called a distributive lattice if for any $a, b, c \in L$

$$a * (b \oplus c) = (a * b) \oplus (a * c)$$

$$a \oplus (b * c) = (a \oplus b) * (a \oplus c)$$

Example :

show that the lattices given by the diagram are not distributive.



Solution:

$$a_3 * (a_1 \oplus a_2) = a_3 * 1 = a_3 = (a_3 * a_1) \oplus (a_3 * a_2)$$

$$a_1 * (a_2 \oplus a_3) = 0 = (a_1 * a_2) \oplus (a_1 * a_3)$$

But

$$a_2 * (a_1 \oplus a_3) = a_2 * 1 = a_2$$

$$(a_2 * a_1) \oplus (a_2 * a_3) = 0 * a_3 = a_3$$

hence the lattices is not distributive . In the other case,

$$b_1 * (b_2 \oplus b_3) = b_1 \text{ while } (b_1 * b_2) \oplus (b_1 * b_3) = 0$$

which shows that the lattices is not distributive.

BOOLEAN ALGEBRA

Definition: A lattice which is complemented and distributive is called a Boolean algebra. or

If B is a non empty set with two binary operations $+$ and $\cdot$, two distinct elements 0 and 1 and a unary operation $'$, then B is called a Boolean algebra if the following properties hold for all a, b, c in B :

B1: Identity laws: $a + 0 = a$, $a \cdot 1 = a$.

B2: Commutative laws: $a + b = b + a$, $a \cdot b = b \cdot a$

B3: Associative law: $(a + b) + c = a + (b + c)$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

B4: Distributive law:

$$a + (b \cdot c) = (a + b) \cdot (a + c)$$

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

B5: Complement law: $a + a' = 1$, $a \cdot a' = 0$

Example Let $B = \{0, 1\}$ and let $+$, $\cdot$ be two operations in B defined by the following operation tables (a) and (b):

$+$	1	0
1	1	1
0	1	0

(a)

$\cdot$	1	0
1	1	0
0	0	0

(b)

Suppose that the complements are defined by $1^1 = 0$ and $0^1 = 1^1$, then B is a Boolean algebra.

Properties of Boolean Algebra:

1. **Idempotent laws:** $a + a = a$ and $a \cdot a = a$ for all $a \in B$.

Proof:

$a + a = (a + a) \cdot 1$	Identity
$= (a + a) \cdot (a + a')$	Complement
$= a + a \cdot a^1$	Distribution
$= a + 0$	Complement
$= a$	Identity
$a \cdot a = a \cdot a + 0$	Identity
$= a \cdot a + a \cdot a'$	Complement
$= a \cdot (a + a')$	Distribution
$= a \cdot 1$	Complement
$= a$	Identity

2. Dominance laws: $a + 1 = 1$ and $a \cdot 0 = 0$, for all $a \in B$

Proof:
$$\begin{aligned} a + 1 &= (a + 1) \cdot 1 \\ &= (a + 1) \cdot (a + a') \\ &= a + 1 \cdot a' \\ &= a + a' \\ &= 1 \end{aligned}$$

$$\begin{aligned} a \cdot 0 &= a \cdot 0 + 0 \\ &= a \cdot 0 + a \cdot a' \\ &= a \cdot (0 + a') \\ &= a \cdot (a' + 0) \\ &= a \cdot a' = 0 \end{aligned}$$

3. Absorption laws: $a + (a \cdot b) = a$, and $a \cdot (a + b) = a$, for all $a, b \in B$

Proof:

$$\begin{aligned} a + (a \cdot b) &= a \cdot 1 + (a \cdot b) \\ &= a \cdot (1 + b) \\ &= a \cdot 1 \\ &= a \end{aligned}$$

$$\begin{aligned} a \cdot (a + b) &= (a + 0) (a + b) \\ &= a + 0 \cdot b \\ &= a + 0 \\ &= a \end{aligned}$$

4. **De Morgan's laws:** $(a + b)' = a' \cdot b'$ and $(a \cdot b)' = a' + b'$, for all $a, b \in B$.

Proof: The theorem is proved if we show that

$$(a + b) + (a' \cdot b') = 1 \text{ and } (a + b) \cdot (a' \cdot b') = 0$$

$$\begin{aligned}(a + b) + (a' \cdot b') &= b + a + (a' \cdot b') \\&= b + (a + a') \cdot (a + b') \\&= b + 1 \cdot (a + b') \\&= b + a + b' \\&= b + b' + a \\&= 1 + a \\&= 1\end{aligned}$$

$$\begin{aligned}(a + b) \cdot (a' \cdot b') &= ((a + b) \cdot a') \cdot b' \\&= ((a \cdot a') + (b \cdot a')) \cdot b' \\&= (0 + (b \cdot a')) \cdot b' \\&= (b \cdot a') \cdot b' \\&= (b \cdot b') \cdot a' \\&= 0 \cdot a' \\&= 0\end{aligned}$$

$$(a + b') = a' \cdot b'$$

In the similar manner, we can prove the other part.

5. Involution law or Double complement law:

$$(a')' = a \text{ for all } a \in B.$$

Proof:

$$a + a' = 1 \text{ and } a \cdot a' = 0$$

$$\text{i.e., } a' + a = 1 \text{ and } a' \cdot a = 0$$

Hence, a is the complement of a'

Dual and Principle of duality

The dual of any statement in a Boolean algebra B is the statement obtained by interchanging the operations $+$ and $\cdot$, and interchanging the elements 0 and 1 in the original statement.

Example: The dual of $a + a \cdot (b + 1) = a$ is $a \cdot a + (b \cdot 0) = a$.

Subalgebra: If C is a non-empty subset of a Boolean algebra such that C itself is a Boolean algebra with respect to the operations in B , then C is called a subalgebra of B .

Boolean Homomorphism:

If $\{B, +, \cdot, ', 0, 1\}$ and $\{C, \cup, \cap, -, \alpha, \beta\}$ are two Boolean algebras, then a mapping $f: B \rightarrow C$ is called a Boolean homomorphism, if for any $a, b \in B$,

$$f(a + b) = f(a) \cup f(b), f(a \cdot b) = f(a) \cap f(b),$$

$$f(a') = \overline{f(a)}, f(0) = \alpha \text{ and } f(1) = \beta,$$

where α and β are the zero and unit elements of C .

Isomorphic Boolean Algebras: Two Boolean algebra B and B' are said to be isomorphic if there is one-to-one correspondence between B and B' with respect to three operations.

Boolean Expressions and Boolean Functions:

A **Boolean expression** in n Boolean variables $x_1, x_2, \dots, x_n$ is a finite string of symbols formed recursively as follows:

1. $0, 1, x_1, x_2, \dots, x_n$ are Boolean expressions.
2. If E_1 and E_2 are Boolean expressions, then $E_1 + E_2$ and $E_1 \cdot E_2$ are also Boolean expressions.
3. If E is a Boolean Expression, E' is also a Boolean expression.

Boolean Function: If $x_1, x_2, \dots, x_n$ are Boolean variables, then function from $f: B^n \rightarrow B\{0,1\}$ is called a Boolean function of degree n .

- Each Boolean expression represents a Boolean function, which is evaluated by substituting the value 0 or 1 for each variable.

Example: The values of the Boolean function $f(a, b, c) = ab + c'$ are displayed in the following truth table:

a	b	c	ab	c'	$ab + c'$
1	1	1	1	0	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	0	1	1
0	1	1	0	0	0
0	1	0	0	1	1
0	0	1	0	0	0
0	0	0	0	1	1

Literals: A Boolean variables or its complement is called literal.

Minterm: A minterm in n Boolean variables is a Boolean product of the n literals in which each literal appears exactly once.

Example: $ab, a'b, ab', a'b'$ forms a complete set of minterms for two variables a and b .

Maxterm: A maxterm in n Boolean variables is a Boolean sum of the n literals in which each literal appears exactly once.

Example: $a + b, a' + b, a + b', a' + b'$ forms a complete set of maxterms for two variables a and b .

Disjunctive Normal form (DNF): When a Boolean function is expressed as a sum of minterms then it is said to be in disjunctive normal form.

Conjunctive Normal form (CNF): When a Boolean function is expressed as a product of maxterms then it is said to be in conjunctive normal form.

Complete DNF (CNF): When a Boolean function in n variables is expressed as a sum of (product of) all the 2^n minterms (maxterms), it is said to be in complete DNF (CNF) .

Canonical Form: Boolean functions expressed in the DNF or CNF are said to be in canonical form.

Expression of Boolean function in Canonical form

1. **Truth table method:** If the Boolean function $f(x, y, z)$ is represented by a truth table, we express $f(x, y, z)$ in DNF as follows

x	y	z	f
1	1	1	0
1	1	0	1
1	0	1	1
1	0	0	1
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	0

the required DNF is : $xyz' + xy'z + xy'z'$

CNF: The CNF of f is the Boolean product of these maxterms.

$$f = (x' + y' + z')(x + y' + z')(x + y' + z)(x + y + z')(x + y + z)$$

2. **Algebraic method:** To get the DNF of a given Boolean functions, we express it as a sum of products.

Example:

$$\begin{aligned} f &= xy' + xz' = xy'.1 + xz'.1 \\ &= xy' \cdot (z + z') + xz' \cdot (y + y') \\ &= xy'z + xy'z' + xyz' + xy'z' \\ &= xy'z + xy'z' + xyz' \end{aligned}$$

This is the required DNF

Now, let us express the same function in the product of sums

$$\begin{aligned}f &= xy' + xz' \\&= x \cdot (y' + z') = (x + 0) \cdot (y' + z' + 0) \\&= (x + yy') \cdot (y' + z' + xx') \\&= (x + y) \cdot (x + y') \cdot (y' + z' + x) \cdot (y' + z' + x') \\&= (x + y + zz') \cdot (x + y' + zz') \cdot (y' + z' + x) \cdot (y' + z' + x') \\&= (x + y + z) \cdot (x + y + z') \cdot (x + y' + z) \cdot (x + y' + z') \\&\quad \cdot (y' + z' + x) \cdot (y' + z' + x') \\&= (x + y + z) \cdot (x + y + z') \cdot (x + y' + z) \cdot (x + y' + z') \\&\quad \cdot (y' + z' + x')\end{aligned}$$

This is the CNF